

Question Paper

Exam Date & Time: 22-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH, MAKEUP EXAMINATIONS - DECEMBER 2025

Computer Vision-III [CSE_4031]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A biometric authentication system uses facial images to verify a person's identity. Explain the steps in digital image processing that help to recognize facial features accurately. (5)
A)

Course Outcome: "CO1" / Cognitive Level: "Understand"
- B) Analyze Log Transformation to enhance the visibility of darker regions in images with large intensity variations. Discuss with the help of a suitable diagram (5)

Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) Illustrate how Laplacian edge detection technique help in obtaining important structural details from digital images. (5)
A)

Course Outcome: "CO2" / Cognitive Level: "Understand"
- B) Apply the Scale-Invariant Feature Transform technique to detect and analyse abnormal regions in medical images and discuss the steps in detail. (5)

Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) In a satellite system, images of the same area are taken at different times and angles, but they don't perfectly overlap due to changes in position and viewing angle. Make use of geometric transformations to align these multi-angle images accurately for precise environmental monitoring and change detection (5)
A)

Course Outcome: "CO3" / Cognitive Level: "Application"
- B) A wildlife drone tracks animals, but due to the camera's tilt and changing flight angles, it often misjudges their actual ground positions. Apply camera calibration to explain how determining intrinsic and extrinsic parameters helps the drone to accurately estimate real-world animal locations and improve tracking precision. (5)

Course Outcome: "CO3" / Cognitive Level: "Application"
- 4) Apply dense motion estimation to generate motion vectors for every pixel in heart motion analysis and explain in detail (5)
A)

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) A traffic surveillance system must separately track multiple vehicles moving at different speeds within the same frame. Model how learned motion estimation groups pixels into motion-consistent regions to track multiple moving objects simultaneously. (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Apply a machine learning-based approach to create an underwater object detection and species classification system using submarine camera images. Describe how the identification of marine species can be done in accurate way. (5)
- A)

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Apply a computer vision based activity detection model to develop a child safety monitoring system that detects signs of abuse using indoor camera footage. How would you apply these techniques to automatically identify normal vs. abnormal activities and trigger alerts in real-time. (5)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 CSE_4031:Computer Vision-III

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 CSE_4031:Computer Vision-III

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	10(20%)
3	Application	7	35(70%)
4	Analysis	1	5(10%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)